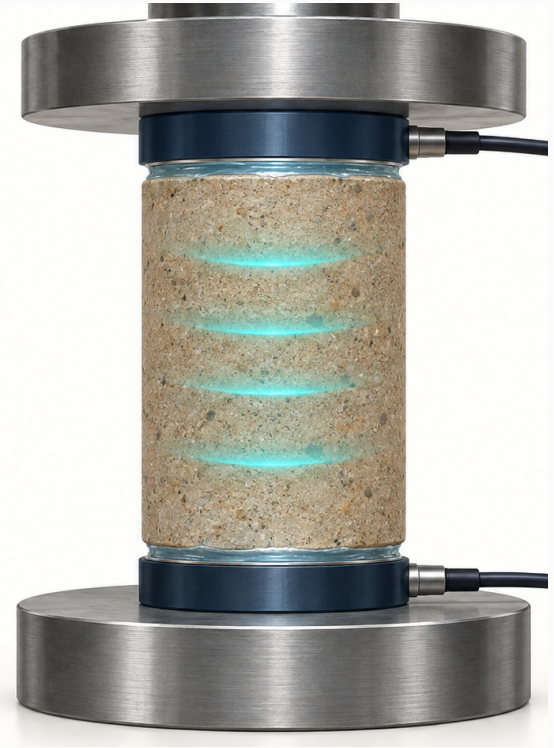
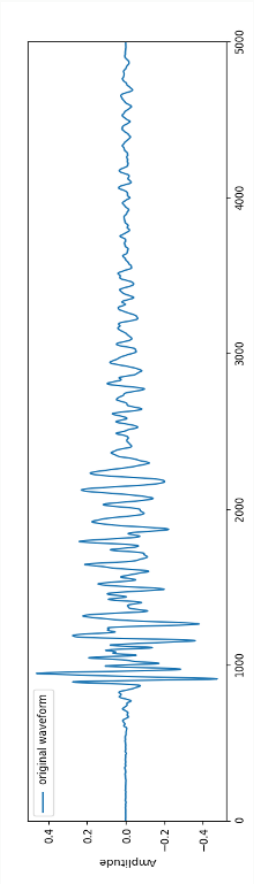
Elastic waves

From earthquakes to laboratory ultrasonics

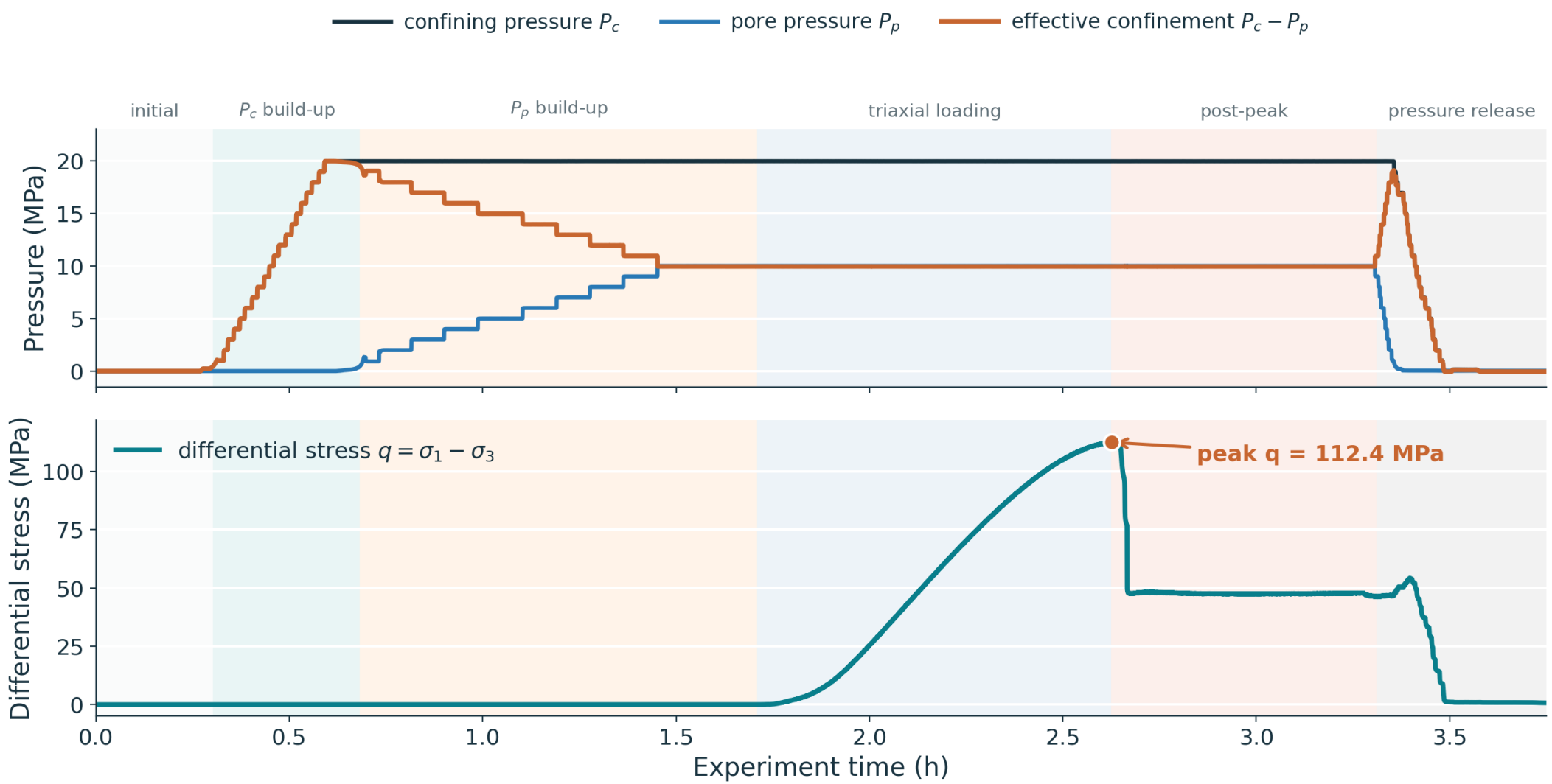
How can a wave reveal something that we cannot see inside a rock?





Why real rocks depart from the ideal model ?

b849: complete pressure and differential-stress history



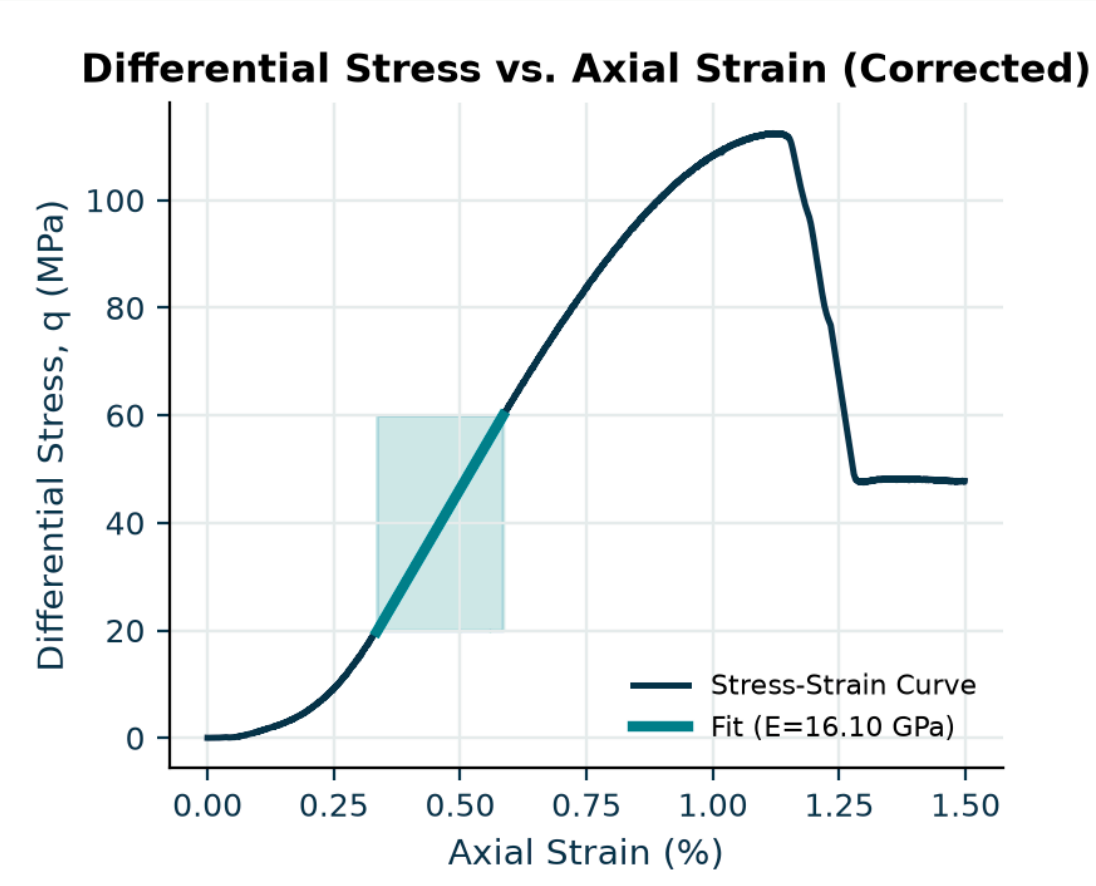
Paterson & Wong (2005) §5.3, pp. 70–71; §5.4.1, pp. 76–77

Why real rocks depart from the ideal model

Mineral composition influences elastic response, but pores, cracks and their orientations also matter. The same rock can show different apparent stiffness as its stress state changes.

Initial crack closure can produce a curved loading segment. A later, approximately linear interval may provide a useful elastic reference.

At **higher stresses**, damage can change the response again. Detailed damage mechanisms belong to the following part of the course.



Paterson & Wong (2005) §5.3, pp. 70–71; §5.4.1, pp. 76–77

Choosing a stress interval changes the apparent modulus

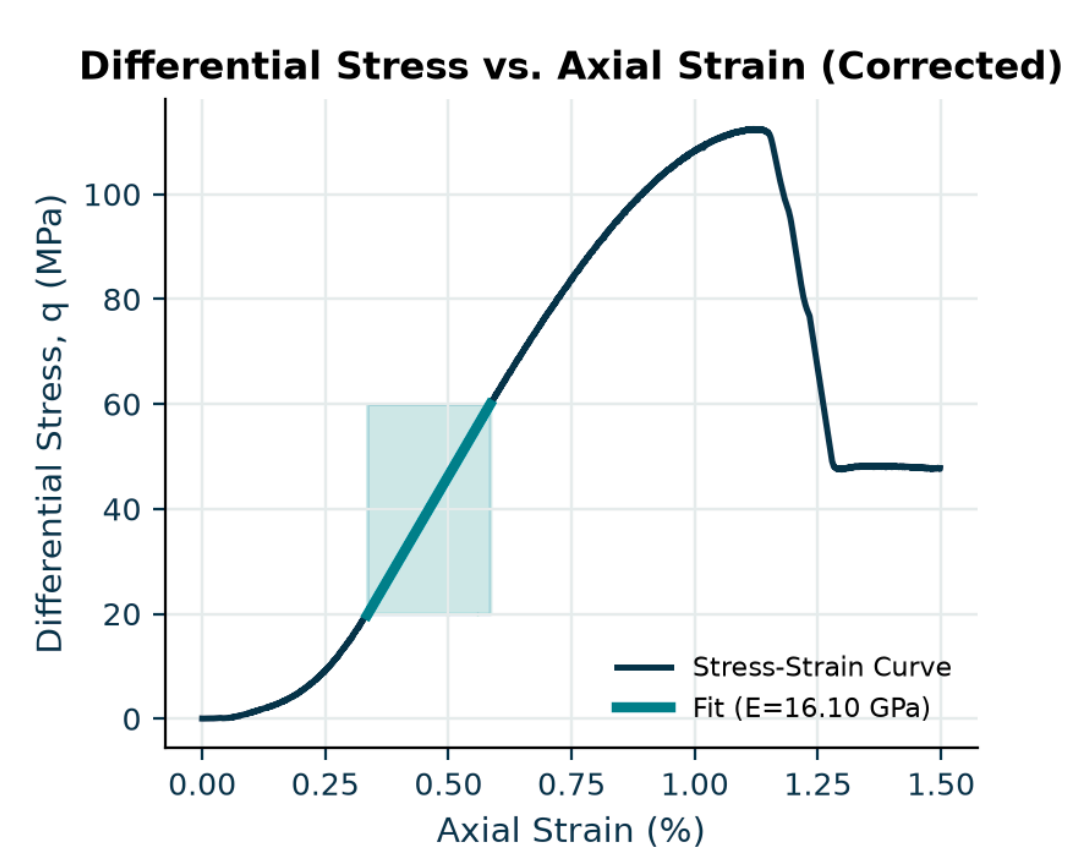
Estimate E from the slope over a stated stress interval. A secant from the origin, a local tangent and an unloading slope need not give the same value.

$$E_{\text{interval}} = \frac{\sigma_{a,2} - \sigma_{a,1}}{\varepsilon_{a,2} - \varepsilon_{a,1}}$$

For the illustrated dataset:

0–20 MPa: about 30.8 GPa

20–60 MPa: 50.0 GPa

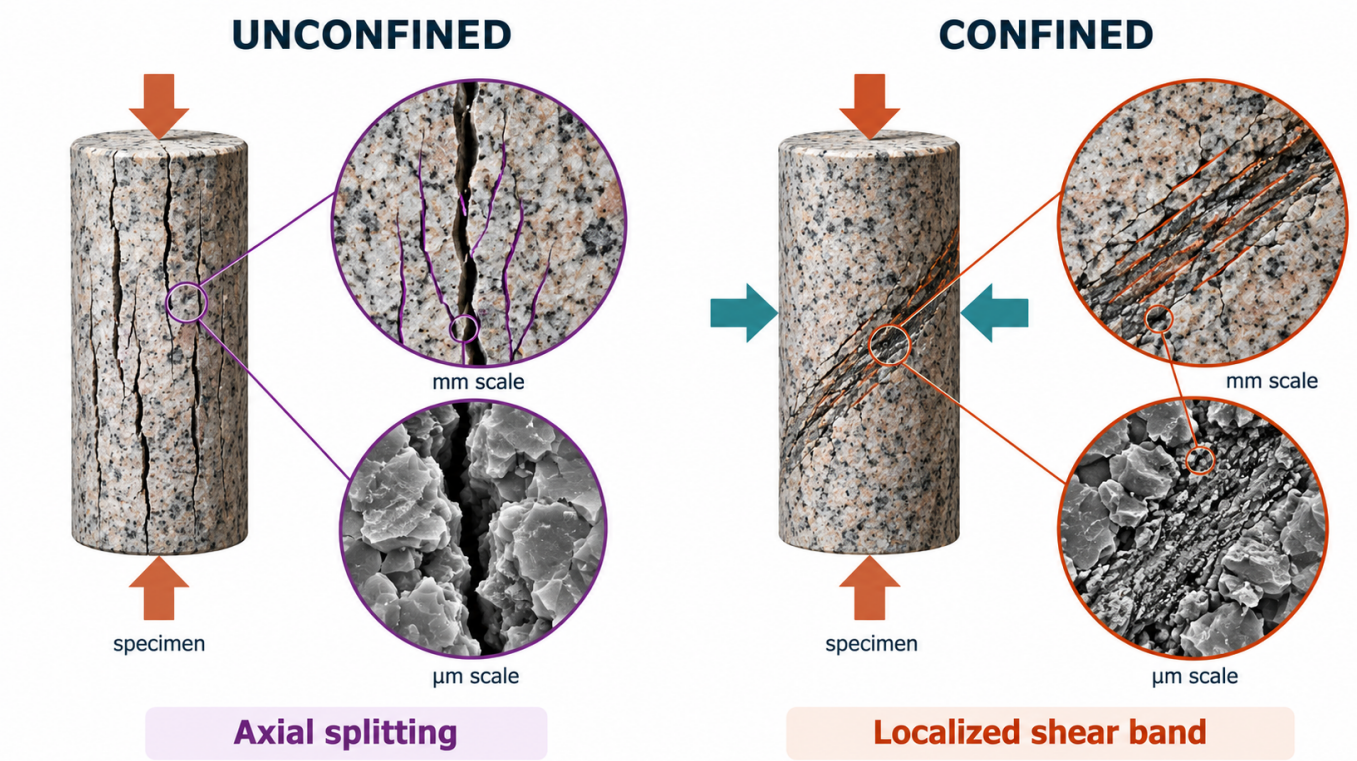


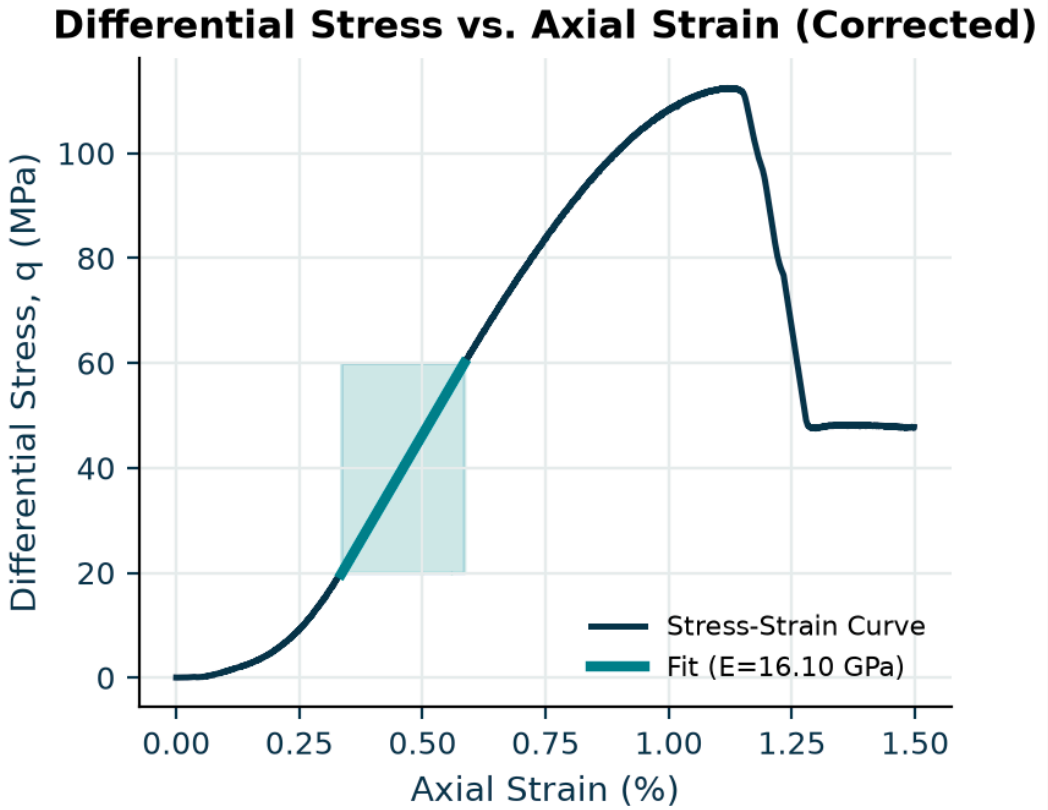
Which interval would you report, and what would unloading help establish?

Activity 1

Paterson & Wong (2005) §5.3, pp. 70–71

Choosing a stress interval changes the apparent modulus





$$E_{\text{interval}} = \frac{\sigma_{a,2} - \sigma_{a,1}}{\varepsilon_{a,2} - \varepsilon_{a,1}}$$

Paterson & Wong (2005) §5.3, pp. 70–71

Learning objectives and lecture sequence

By the end of this lecture, you should be able to:

Learning objective	Main evidence or calculation
Connect field seismology with laboratory ultrasonics	Source–medium–receiver comparison
Explain how piezoelectric transducers transmit and receive	Electrical–mechanical conversion
Distinguish P and S waves	Particle motion and restoring stiffness

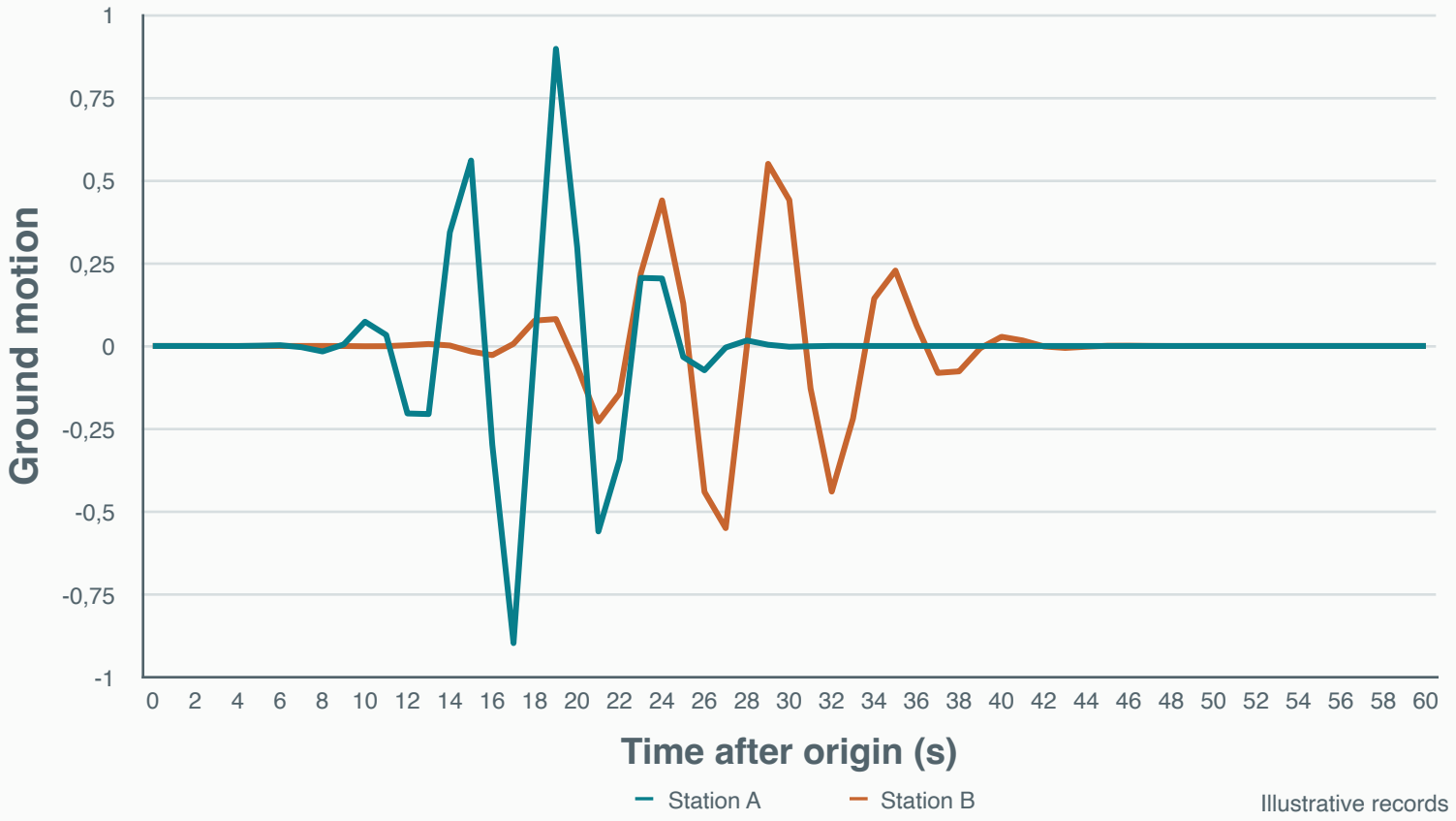
Sequence

Natural observation → laboratory measurement → wave theory → travel time → velocity

Opening question: the same earthquake, two different records

Two stations lie at the same epicentral distance. Their first arrivals and amplitudes differ.

What could differ between the two propagation paths?



Think individually for 30 seconds, then compare one explanation with a neighbour.

A recorded waveform combines three influences



earthquake mechanism
source spectrum

velocity structure
attenuation and scattering

sensor response
local coupling

Recorded signal = source ⊗ path ⊗ receiver

Interpreting the medium requires constraints on the source and receiver.

Seismology turns arrival times into information about the Earth



known or located source

unknown structure

measured motion

Measured

- arrival time,
- amplitude,
- frequency content

Inferred

- velocity structure,
- interfaces,
- attenuation

The laboratory experiment asks the same inverse question at centimetre scale.

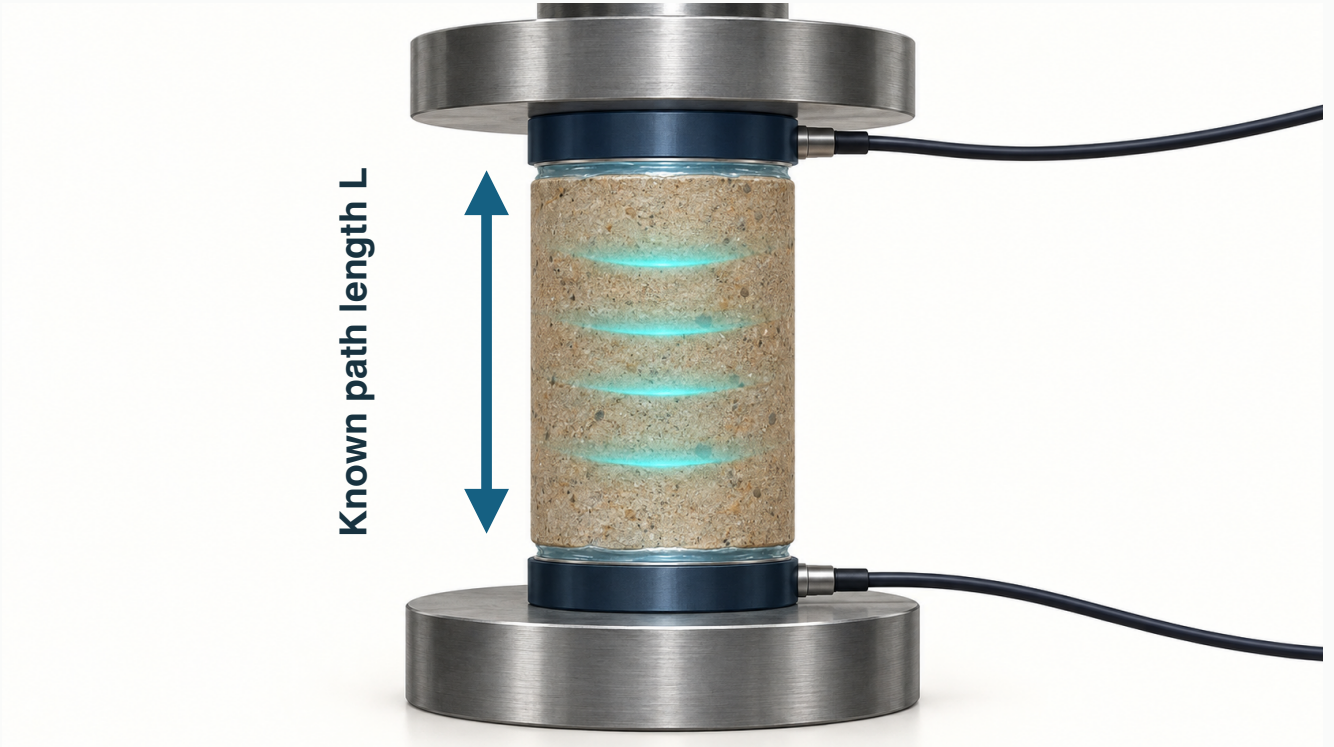
The same experiment can be performed at two scales

Natural seismology	Laboratory ultrasonics
Earthquake or controlled source	Electrical pulse and transmitting transducer
Rock volume in the Earth	Rock specimen
Seismic station	Receiving piezoelectric transducer
Seismogram	Digitized voltage waveform
Propagation over kilometres	Propagation over centimetres
Typical frequencies in Hz	Typical frequencies in kHz–MHz

Source → medium → receiver

Which part of the laboratory experiment do we control most precisely?

Pulse transmission sends a controlled disturbance through the specimen

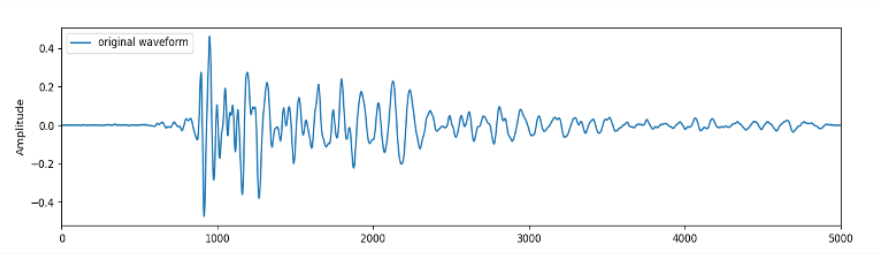


TRANSMITTER (Tx)

electrical pulse produces
a short deformation

RECEIVER (Rx)

deformation produces
a voltage waveform



The measurement chain converts energy twice



Controlled

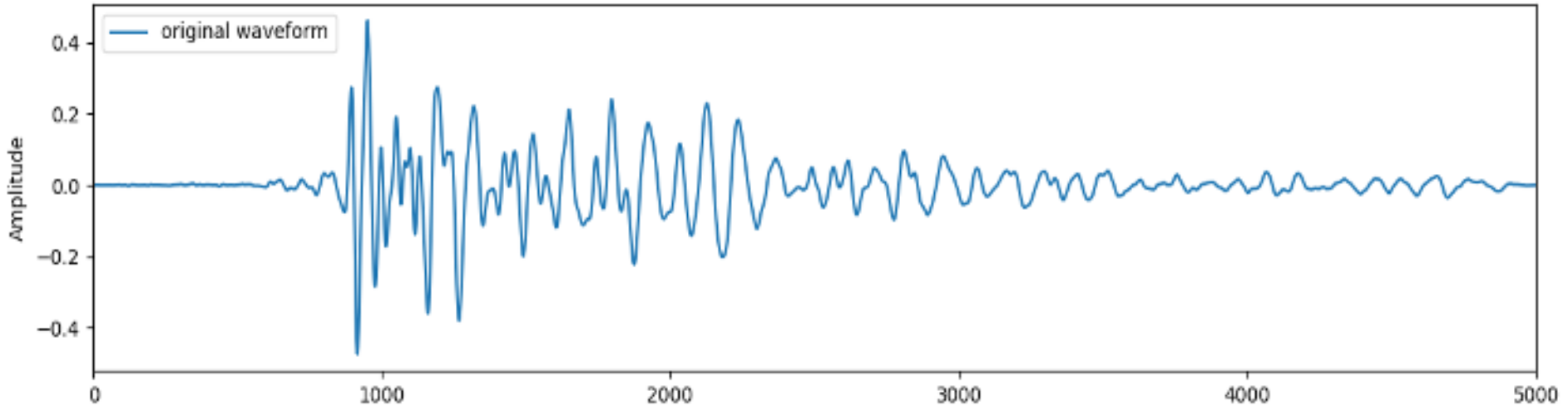
pulse settings, geometry, loading conditions

Measured

voltage waveform sampled at known times

At which steps can experimental artefacts enter the signal?

The instrument records voltage as a function of time



DIRECT OBSERVATION

digitized voltage versus time

INTERPRETATIVE CHOICES

- arrival pick
- wave type
- signal window

The waveform is measured. Travel time and velocity require additional choices.

Question: how does electricity create a mechanical wave?

A voltage pulse enters a solid ceramic disk. A mechanical disturbance leaves the disk and enters the rock.

ELECTRICAL ? **MECHANICAL**

Predict what must happen inside the transducer when the voltage changes.

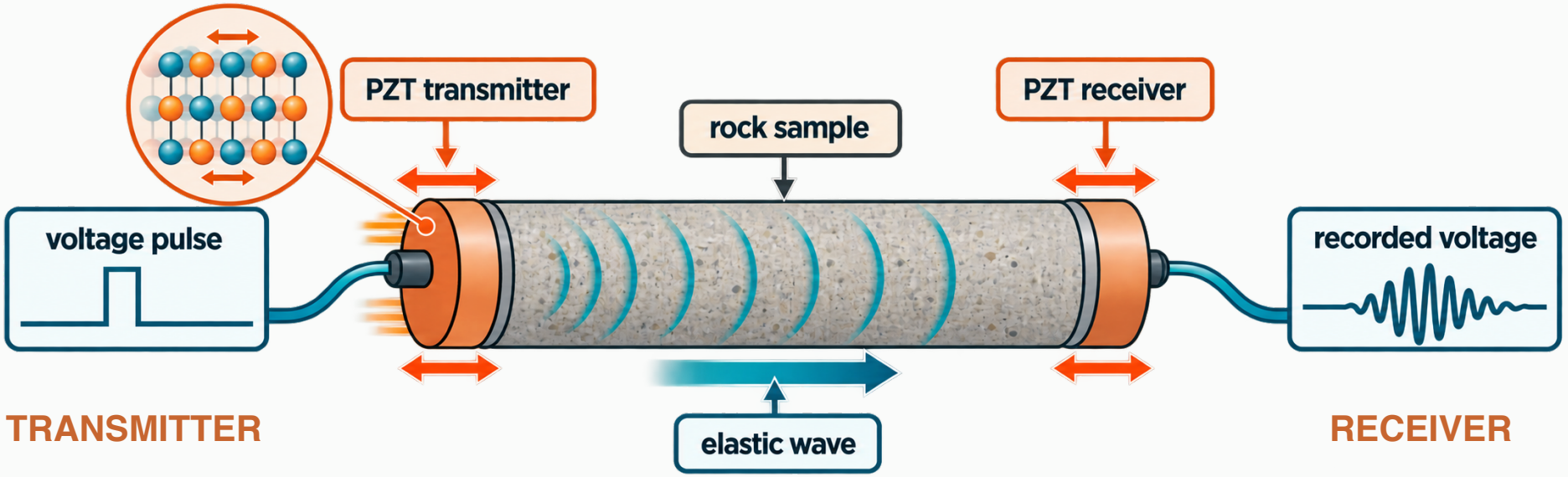
Piezoelectricity couples electrical and mechanical states

INVERSE PIEZOELECTRIC EFFECT

applied voltage → change in shape

DIRECT PIEZOELECTRIC EFFECT

applied stress → electrical charge



The same piezoelectric material can perform both conversions.

The transducer does not measure velocity. It converts deformation into voltage.

Coupling and bandwidth shape the recorded pulse

COUPLING

The contact layer transfers motion across the transducer–rock interface. Its thickness and contact pressure affect amplitude and repeatability.

BANDWIDTH

The transducer favours a frequency band. The emitted pulse rings for several cycles and cannot reproduce every frequency equally.

ALIGNMENT

Tilt, uneven pressure or changing contact during loading can alter the waveform even when the rock remains unchanged.

Consequences for interpretation

Travel time is often more repeatable than amplitude. Both require stable geometry and calibration.

A waveform change can originate in the rock, the transducers or their contacts.

Learning check: the waveform changes between two identical runs

The specimen has not been reloaded or moved. The first arrival occurs at the same time, but the peak amplitude decreases by 30%.

Which conclusion is justified?

A. The rock developed new cracks

B. The rock velocity decreased

C. The signal changed, but the cause remains ambiguous

D. The propagation distance increased

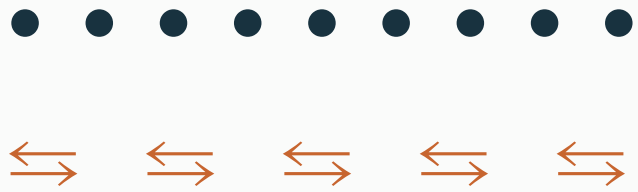
Vote individually, then defend your choice using the measurement chain.

An elastic wave transports a disturbance through the material

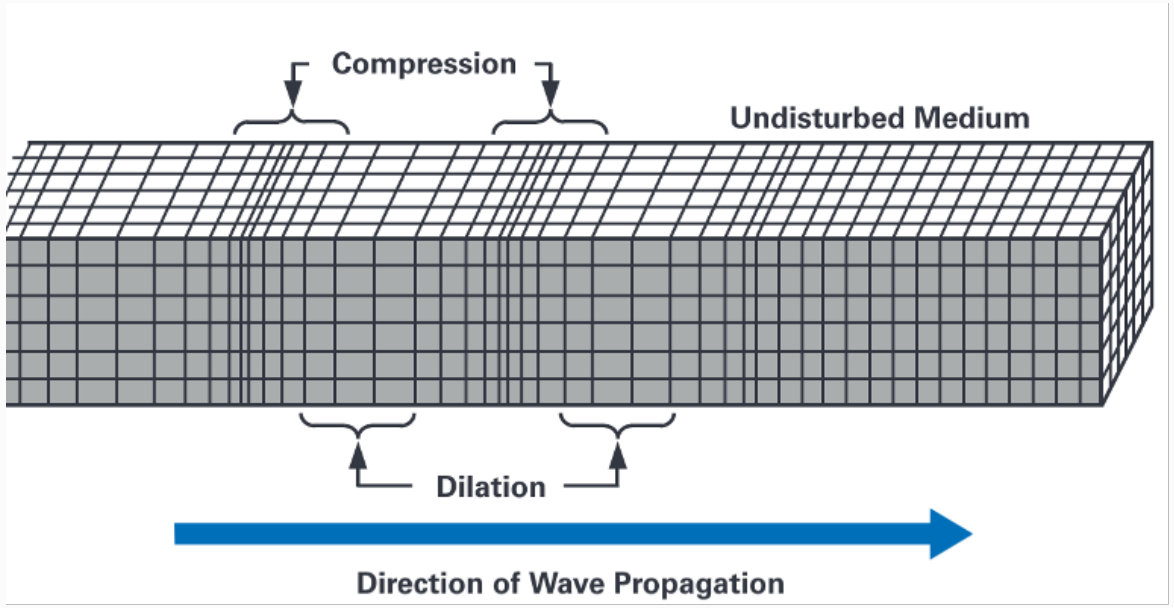
Particles oscillate around their equilibrium positions.

The disturbance propagates through **elastic interaction** between neighbouring particles.

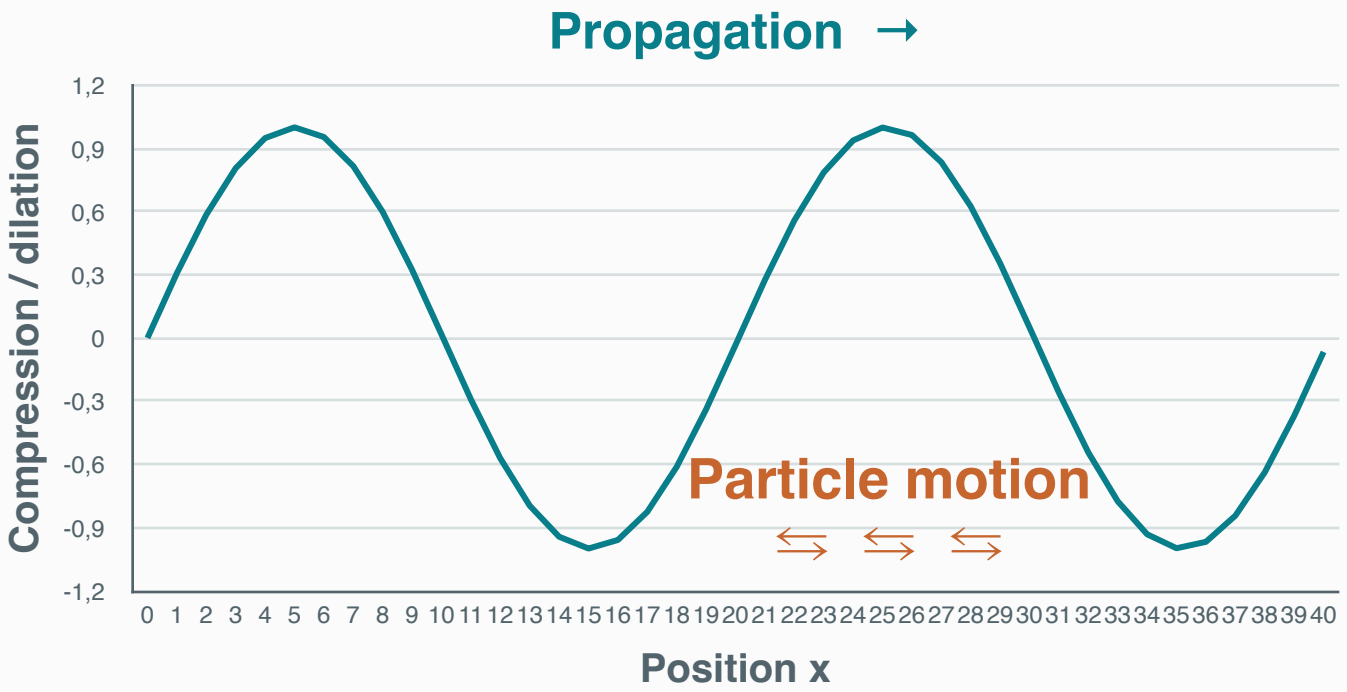
Matter does not travel from transmitter to receiver.

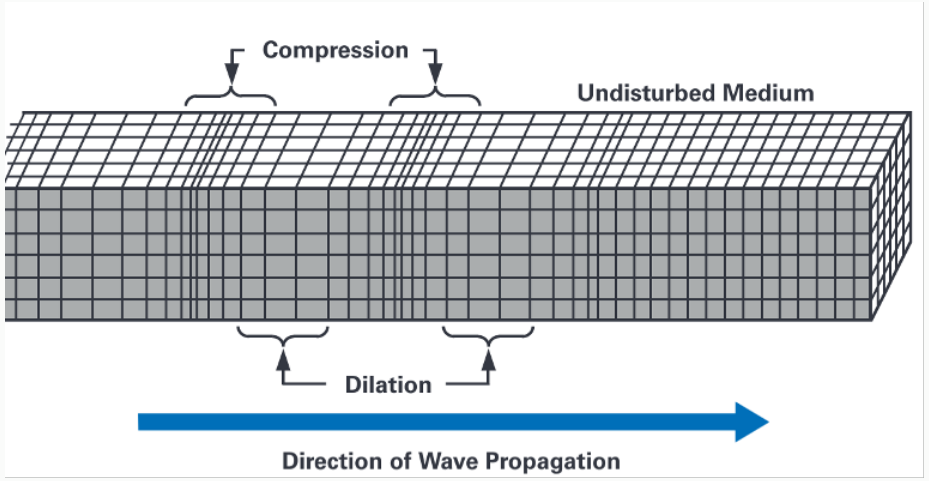


local oscillation



P waves move particles parallel to propagation

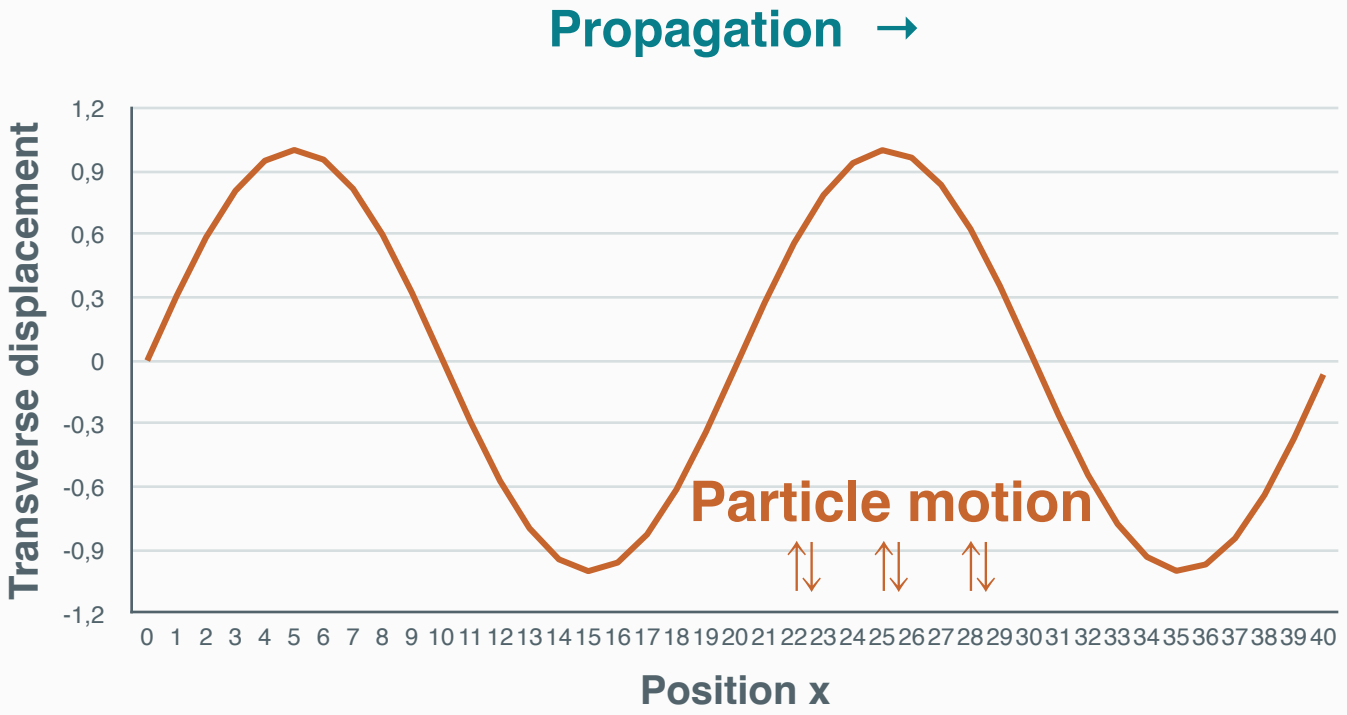


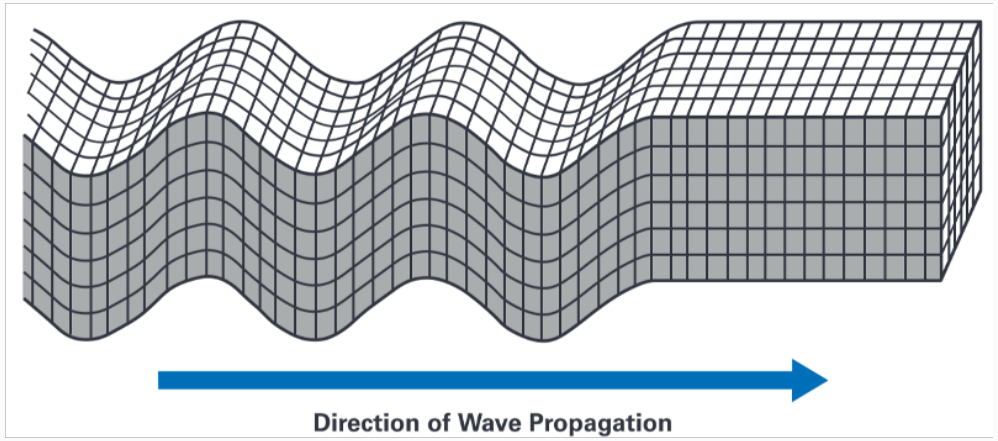


Alternating compression and dilation change volume locally.

P means primary: in ordinary rocks, this wave arrives first.

S waves move particles perpendicular to propagation





Shear deformation changes shape without first-order volume change.

S means secondary: it arrives after the P wave.

Wave speed increases with stiffness and decreases with density

$$V \propto \sqrt{(\text{stiffness} / \text{density})}$$

P WAVE

$$V_p = \sqrt{\frac{K + \frac{4}{3}G}{\rho}}$$

resists compression and shear

S WAVE

$$V_s = \sqrt{\frac{G}{\rho}}$$

resists shear

K and G were introduced in Lecture 3. Their inversion from V_p and V_s belongs to Lecture 5.

Learning check: which statements must be true?

Assume a homogeneous, isotropic, linear elastic solid with positive K and G.

1. V_p must exceed V_s.

2. Increasing density while keeping the moduli fixed increases both speeds.

3. A pure fluid can transmit an S wave.

4. P-wave motion is parallel to propagation.

$$V_s = \sqrt{\frac{G}{\rho}}$$

$$V_p = \sqrt{\frac{K + \frac{4}{3}G}{\rho}}$$

Decide individually, then explain each choice using the equations or particle motion.

What we can now measure

OBSERVE	CALCULATE
voltage waveform versus time	travel time wave velocity

Next lecture

How do V_p, V_s and waveform changes constrain dynamic moduli, cracks, fluids, stress and damage?

Feedback welcome

Reading: Paterson & Wong (2005), §5.4, pp. 76–88. Focus on what elastic-wave speed and attenuation reveal about cracks.